

JEE MAIN REVISION MASTER: PARABOLA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Standard Parabola, Focus, Directrix & Tangents	3 PYQs	Parabola
Standard	Normals to Parabola & Co-Normal Points	0 PYQs	Normals

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Standard Parabola, Focus, Directrix & Tangents	$y^2 = 4ax \implies \text{Focus} (a, 0), \text{Directrix: } x = -a, \text{LR} = 4a, \text{Tangent: } y = mx + \frac{a}{m}, \text{Parametric: } (at^2, 2at)$
Normals to Parabola & Co-Normal Points	$\text{Normal: } y = mx - 2am - am^3, \text{Parametric: } y + tx = 2at + at^3, \sum m_i = 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Standard Parabola, Focus, Directrix & Tangents (Moderate Yield — 3 PYQs)

Core Principle: Focal properties of $y^2 = 4ax$, slope form of tangent $y = mx + a/m$, and parametric chords.

■ **Caution / JEE Pitfall:** For $x^2 = 4ay$, the tangent in slope form is $y = mx - am^2$, NOT $y = mx + a/m$!

■ **Speed Trick:** The tangents at the extremities of any focal chord intersect at right angles (90°) strictly on the DIRECTRIX.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The equation of parabola whose focus is (6,0) and directrix is $x=-6$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the parabola $y=\alpha x^2-6x+\beta$ passes through the point (0,2) and has its tangent at $x=(3/2)$ parallel to X-axis, the..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "If the parabola $x^2=4ay$ passes through the point (2,1), then the length of the latus rectum is..." [Ans: B]

3.2 Normals to Parabola & Co-Normal Points (Standard Yield — 0 PYQs)

Core Principle: Cubic normal equation $am^3 + (2a - x)m + y = 0$, maximum of three normals from external point, and co-normal properties.

■ **Caution / JEE Pitfall:** From an external point (h, k) , three normals can be drawn to $y^2 = 4ax$ if and only if $h > 2a$.

■ **Speed Trick:** The algebraic sum of the slopes of three co-normal points is always ZERO: $m_1 + m_2 + m_3 = 0$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The equation of parabola whose focus is $(6,0)$ and directrix is $x=-6$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the parabola $y = \alpha x^2 - 6x + \beta$ passes through the point $(0,2)$ and has its tangent at $x = (3/2)$ parallel to X-axis, the..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The equation of parabola whose focus is $(6,0)$ and directrix is $x=-6$ is

- (A) $y^2 = 24x$
- (B) $y^2 = -24x$
- (C) $x^2 = 24y$
- (D) $x^2 = -24y$

Correct Answer: None

Solution Analysis: ■ Focus has positive x-coordinate. So, coordinate of focus $= (a,0) \equiv (6,0) \therefore a=6$ (i) Hence, equation of parabola is $y^2 = 24x$ [■ $y^2 = 4ax$ and using Eq. (i)]